Prayas JEE AIR (2025)

WEEKLY TEST 34

DURATION : 180 Minutes | **DATE : 20/01/2025** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25** questions in each part in which first **20** questions are of Objective Type and Last **5** questions are integers type and all **25** questions are **compulsory.**
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number: _____

OMR Bar Code Number: _____

Candidate's Signature: _____ **Invigilator's Signature** _____

SECTION-I (PHYSICS)

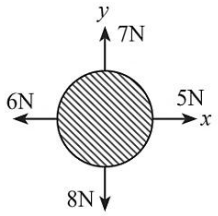
Single Correct Type Questions

1. A ball is thrown up vertically with a certain velocity at $t = 0$ so that, it reaches a maximum height h. Find the ratio of the time instants in which it is at height $\frac{h}{3}$ while going up and coming down respectively.

(1) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$ (2) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

(3) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (4) $\frac{1}{3}$

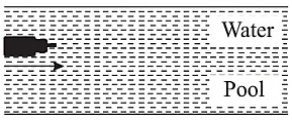
2. For a free body diagram shown in the figure, the four forces are applied in the 'x' and 'y' directions. What additional force must be applied and at what angle with positive x-axis so that the net acceleration of body is zero?



(1) $\sqrt{2}\,\text{N}, 45^\circ$ (2) $\sqrt{2}\,\text{N}, 135^\circ$

(3) $\frac{2}{\sqrt{3}}\text{N}, 30^\circ$ (4) $2\,\text{N}, 45^\circ$

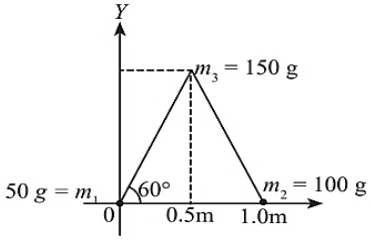
3. A bullet of mass 200g having initial kinetic energy 90 J is shot inside a long swimming pool as shown in the figure. If its kinetic energy reduces to 40 J within 1 s, the minimum length of the pool, the bullet has to travel so that it completely comes to rest is



(1) 45 m (2) 90 m

(3) 125 m (4) 25 m

4. Three particles of masses 50 g, 100 g and 150 g are placed at the vertices of an equilateral triangle of side 1 m (as shown in the figure). The (x, y) coordinates of the centre of mass of system of these three particles will be :



(1) $\left(\frac{\sqrt{3}}{7}\text{m}, \frac{7}{12}\text{m}\right)$ (2) $\left(\frac{7}{12}\text{m}, \frac{\sqrt{3}}{8}\text{m}\right)$

(3) $\left(\frac{\sqrt{3}}{4}\text{m}, \frac{5}{12}\text{m}\right)$ (4) $\left(\frac{7}{12}\text{m}, \frac{\sqrt{3}}{4}\text{m}\right)$

5. Match **Column**–I with **Column**–II

Choose the correct answer from the options given below:

	Column-I		**Column-II**
A	Moment of inertia of solid sphere of mass M and radius R about any tangent.	P	$\frac{5}{3}MR^2$
B	Moment of inertia of hollow sphere of mass M and radius R about any tangent	Q	$\frac{7}{5}MR^2$
C	Moment of inertia of circular ring of mass M and radius R about its diameter.	R	$\frac{1}{4}MR^2$
D	Moment of inertia of circular disc of mass M and radius R about any diameter.	S	$\frac{1}{2}MR^2$

(1) A→Q; B→P; C→S; D→R

(2) A→P; B→Q; C→S; D→R

(3) A→Q; B→P; C→R; D→S

(4) A→P; B→Q; C→R; D→S

6. A double star is a system of two stars moving around the centre of mass of the system due to mutual gravitational force. Find the distance between the components of the double star system if the total mass equals M and time period is T.

(1) $d = \frac{\left(GMT^2\right)^{\frac{1}{3}}}{4\pi^2}$ (2) $d = \left(\frac{GMT^2}{4\pi^2}\right)^{\frac{1}{3}}$

(3) $d = \frac{\left(GMT^2\right)^{\frac{1}{3}}}{2\pi}$ (4) $d = \left(\pi GMT^2\right)^{\frac{1}{3}}$

7. Match Column-I with Column -II

Column-I		**Column-II**	
A	Torque	P	$ML^{-2}T^{-2}$
B	Stress	Q	ML^2T^{-2}
C	Pressure gradient	R	$ML^{-1}T^{-1}$
D	Coefficient of viscosity	S	$ML^{-1}T^{-2}$

Choose the correct answer from the options given below:

(1) A→R, B→S, C→P, D→Q

(2) A→S, B→Q, C→R, D→P

(3) A→Q, B→S, C→P, D→R

(4) A→Q, B→P, C→S, D→R

8. The area of cross section of the rope used to lift a load by a crane is 2.5×10^{-4} m^2. The maximum lifting capacity of the crane is 10 metric tons. To increase the lifting capacity of the crane to 25 metric tons, the required area of cross section of the rope should be: (take g = 10 ms^{-2})

(1) 6.25×10^{-4} m^2 (2) 10×10^{-4} m^2

(3) 1×10^{-4} m^2 (4) 1.67×10^{-4} m^2

9. A free neutron decays into a proton but a free proton does not decay into neutron. This is because

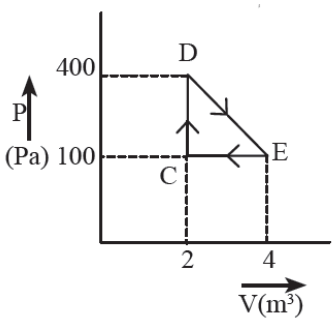
(1) neutron is an uncharged particle

(2) proton is a charged particle

(3) neutron is a composite particle made of a proton and an electron

(4) neutron has larger rest mass than proton

10. A thermodynamic system is taken through cyclic process. The total work done by the gas in the cyclic process is:



(1) 100 J (2) 300 J

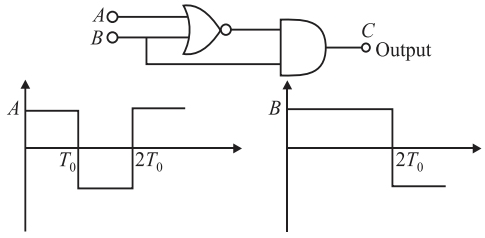
(3) Zero (4) 200 J

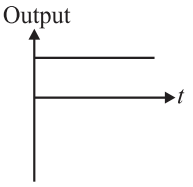
11. Assume that a tunnel is dug along a chord of the earth, at a perpendicular distance (R/2) from the earth's centre, where 'R' is the radius of the Earth. The wall of the tunnel is frictionless. If a particle is released in this tunnel, it will execute a simple harmonic motion with a time period :

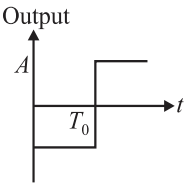
(1) $2\pi\sqrt{\frac{R}{g}}$ (2) $2\pi\sqrt{\frac{2R}{g}}$

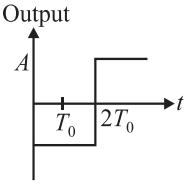
(3) $\frac{\pi}{2}\sqrt{\frac{g}{R}}$ (4) $2\pi\sqrt{\frac{R}{2g}}$

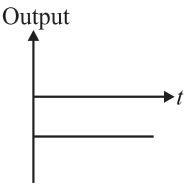
12. Correct output for given logic circuit and inputs is:



(1)


(2)


(3)


(4)


13. An electron of a hydrogen like atom, having Z = 4, jumps from 4th energy state to 2nd energy state, The energy released in this process, will be:
(Given Rch = 13.6 eV)
Where R = Rydberg constant
c = Speed of light in vacuum
h = Planck's constant

(1) 54.4 eV (2) 27.2 eV

(3) 40.8 eV (4) 68.0 eV

14. A parallel plate capacitor C_0 is charged by a battery of constant emf V_0.
Case-I: The energy stored in the capacitor while the battery is kept connected and the plate separation is doubled is U_1.
Case-II: The energy stored in the capacitor when the charging battery is disconnected and the separation between plates is doubled is U_2.
Then find $\frac{U_1}{U_2}$

(1) 4 (2) $\frac{1}{4}$

(3) 2 (4) $\frac{1}{2}$

15. A galvanometer together with an unknown resistance in series is connected across a set of two identical batteries each of 1.5V. When the batteries are connected in series, the galvanometer records a current of 1A, and when the batteries are in parallel the current is 0.6A. What is the internal resistance of the battery?

(1) 0.66 Ω (2) 0.25 Ω

(3) 0.33 Ω (4) None of these

16. The magnetic field at the centre of a circular coil of radius *r*, due to current *I* flowing through it, is *B*. The magnetic field at a point along the axis at a distance *r*/2 from the centre is:

(1) $\frac{B}{2}$ (2) $2B$

(3) $\left(\frac{2}{\sqrt{5}}\right)^3 B$ (4) $\left(\frac{1}{\sqrt{3}}\right)^3 B$

17. A series *L*-*R* circuit is connected to a battery of emf *V*. If the circuit is switched on at $t = 0$, then the time at which the energy stored in the inductor reaches $\left(\frac{1}{n}\right)$ times of its maximum value, is

(1) $\frac{L}{R}\ln\left(\frac{\sqrt{n}+1}{\sqrt{n}-1}\right)$ (2) $\frac{L}{R}\ln\left(\frac{\sqrt{n}}{\sqrt{n}-1}\right)$

(3) $\frac{L}{R}\ln\left(\frac{\sqrt{n}+2}{\sqrt{n}+1}\right)$ (4) $\frac{L}{R}\ln\left(\frac{2\sqrt{n}+1}{\sqrt{n}}\right)$

18. A 60 Hz AC voltage of 200 V impressed across an *LR*-circuit results in a current of 2 A. If the power dissipation is 320 W, calculate the rms value of the back emf (in volt) arising in the inductance.

(1) 160 V (2) 140 V

(3) 120 V (4) 180 V

19. A plane electromagnetic wave is propagating along the direction $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$, with its polarization (Electric field) along the direction $\hat{k}$. The correct form of the magnetic field of wave would be (here B_0 is an appropriate constant K is angular wave number and ω is angular frequency)

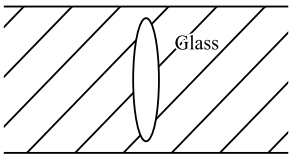
(1) $B_0\frac{\hat{i}+\hat{j}}{\sqrt{2}}\cos\left(\omega t - K\frac{r\hat{i}+r\hat{j}}{\sqrt{2}}\right)$

(2) $B_0\frac{\hat{i}-\hat{j}}{\sqrt{2}}\cos\left(\omega t + K\frac{r\hat{i}+r\hat{j}}{\sqrt{2}}\right)$

(3) $B_0\frac{\hat{i}-\hat{j}}{\sqrt{2}}\cos\left(\omega t - K\frac{r\hat{i}+r\hat{j}}{\sqrt{2}}\right)$

(4) $B_0\hat{k}\cos\left(\omega t - K\frac{r\hat{i}+r\hat{j}}{\sqrt{2}}\right)$

20. There is an air lens in an extended glass medium. The radius of curvature of both the curved surfaces is *R*, and refractive index of the glass is $\frac{3}{2}$. Power of this air lens is *P*. Find the refractive index of the material to be filled inside the lens so that its power becomes –*P*.



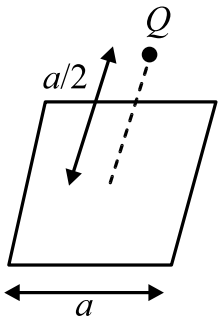
(1) 2 (2) 3

(3) 2.5 (4) $\frac{4}{3}$

Integer Type Questions

21. The width of fringe is 2 mm on the screen in a double slit experiment for the light of wavelength of 400 nm. The width of the fringe (in mm) for the light of wavelength 600 nm will be:

22. A charge *Q* is placed at a distance $\frac{a}{2}$ above the centre of the square surface of edge *a* as shown in the figure. The electric flux through the square surface is $\frac{Q}{x\varepsilon_0}$. Find *x*



23. A bakelite beaker has volume capacity of 500 cc at 30°C. When it is partially filled with V_m volume (at 30°C) of mercury, it is found that the unfilled volume of the beaker remains constant as temperature is varied. If $\gamma_{(beaker)} = 6 \times 10^{-6}\ °C^{-1}$ and $\gamma_{(mercury)} = 1.5 \times 10^{-4}\ °C^{-1}$, where γ is the coefficient of volume expansion, then V_m (in cc) is close to _____.

24. A photoelectric surface is illuminated successively by monochromatic light of wavelength λ and λ/2 if the maximum kinetic energy of the emitted photoelectrons in the second case is 3 times that in the first case. If the work function of the surface of the material is ϕ then find the value of $\left(\frac{hc}{\lambda\phi}\right)$ is

(h = Planck's constant, c = speed of light)

25. An organ pipe P_1 closed at one end vibrating in its first overtone and another pipe P_2 open at both ends vibrating in third overtone are in resonance with a given tuning fork. If the ratio of the length of P_1 to that of P_2 is ϕ than find the value of 8ϕ .

SECTION-II (CHEMISTRY)

Single Correct Type Questions

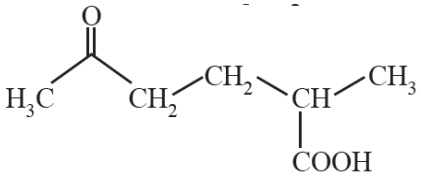
26. The depression in freezing point for 0.1 m Na_2SO_4, 0.2 m urea and 0.05 m $Ca_3(PO_4)_2$ are in the ratio of (Assume 100% ionisation)

(1) 3 : 1 : 5 (2) 2 : 4 : 1
(3) 6 : 4 : 5 (4) 1 : 2 : 5

27. The compound that cannot act as oxidising as well as reducing agent is:

(1) HNO_2 (2) H_3PO_4
(3) H_2SO_3 (4) H_2O_2

28. The correct IUPAC nomenclature for the following compound is

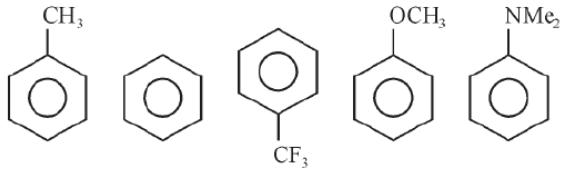


(1) 5–Formyl–2–methylhexanoic acid
(2) 2–Methyl–5–oxohexanoic acid
(3) 2–Formyl–5–methylhexan–6–oic acid
(4) 5–Methyl–2–oxohexan–6–oic acid

29. The crystal field stabilization energy (CFSE) of $[Fe(H_2O)_6]Cl_2$ and $K_2[NiCl_4]$, respectively, are

(1) $-0.4\Delta_o$ and $-0.8\Delta_t$
(2) $-0.4\Delta_o$ and $-1.2\Delta_t$
(3) $-2.4\Delta_o$ and $-1.2\Delta_t$
(4) $-0.6\Delta_o$ and $-0.8\Delta_t$

30. Decreasing order of reactivity towards electrophilic substitution for the following compounds is:



a b c d e

(1) c > b > a > d > e
(2) e > d > a > b > c
(3) a > d > e > b > c
(4) d > a > e > c > b

31. From amongst the following alcohols the one that would react fastest with conc. HCl and anhydrous $ZnCl_2$, is

(1) 2–Butanol
(2) 2–Methylpropan–2–ol
(3) 2–Methylpropanol
(4) 1–Butanol

32. Given the equilibrium constant K_c of the reaction: $Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$ is 10×10^{15}, Calculate the E°_{cell} of this reaction at 298 K $\left[2.303\frac{RT}{F} \text{ at } 298\,K = 0.059\,V\right]$

(1) 0.4736 mV (2) 0.4736 V
(3) 0.04736 V (4) 0.04736 mV

33. Compound A contains 8.7% Hydrogen, 74% Carbon and 17.3% Nitrogen. The molecular formula of the compound is, Given: Atomic masses of C, H and N are 12, 1 and 14 amu respectively. The empirical formula of compound A is

(1) $C_4H_6N_2$ (2) C_2H_3N
(3) C_5H_7N (4) $C_4H_6N_3$

34. In general, the properties that decreases and increase down a group in the periodic table respectively are

(1) Atomic radius and electrongegativity
(2) Electron gain enthalpy and electronegativity
(3) Electronegativity and atomic radius
(4) Electronegativity and electron gain enthalpy

35. For a gaseous reaction, $2A + B_2 \rightarrow 2AB$, the following rate data was obtained at $300^\circ C$.

Rate of disappearance of B_2 (Mole litre^{-1} min^{-1})	**[A] (mole/litre)**	**[B_2] (mole/litre)**
1.8×10^{-3}	0.015	0.15
1.08×10^{-2}	0.09	0.15
5.4×10^{-3}	0.015	0.45

The rate of formation of AB when [A] is 0.02 M and $[B_2]$ is 0.04 M at 300°C is

(1) 1.28×10^{-3} M min^{-1}
(2) 2.56×10^{-3} M min^{-1}
(3) 6.4×10^{-4} M min^{-1}
(4) 1.28×10^{-2} M min^{-1}

36. Consider the following sets of quantum number

I. n = 4, ℓ = 1 II. n = 4, ℓ = 0
III. n = 3, ℓ = 2 IV. n = 3, ℓ = 1

Arrange the energy levels corresponding to the given sets of quantum number in the increasing order of energy for the multielectron species.

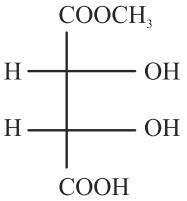
(1) III < IV < II < I (2) IV < II < III < I
(3) II < IV < I < III (4) I < III < II < IV

37. pH of a saturated solution of $Ba(OH)_2$ is 12. Hence, the K_{sp} of the $Ba(OH)_2$ is

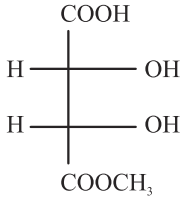
(1) 5×10^{-7} (2) 5×10^{-4}

(3) 10^{-6} (4) 4×10^{-5}

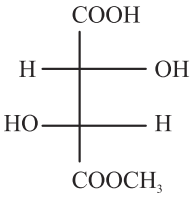
38. The correct statement about the compound A, B and C is



(A)



(B)



(C)

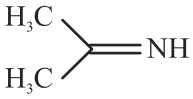
(1) 'A' and 'B' are identical

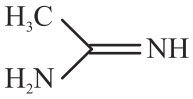
(2) 'A' and 'B' are diastereomers

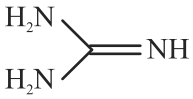
(3) 'A' and 'C' are enantiomers

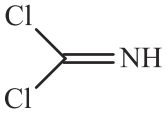
(4) 'A' and 'B' are enantiomers

39. Which among the following compounds is most basic?

(1) 

(2) 

(3) 

(4) 

40. Which is the correct match for all?

List-I		List-II	
A	Sucrose	P	β-(1,4)-Glycosidic linkage
B	Maltose	Q	α, β-(1,2)-Glycosidic linkage
C	Starch	R	natural biopolymer
D	Lactose	S	α-(1,4) Glycoside linkage.

(1) A→S; B→P; C→R,S; D→Q

(2) A→P; B→Q; C→R; D→S

(3) A→Q; B→S; C→R,S; D→P

(4) A→R; B→R,S; C→Q; D→P

41. Arrange following compounds towards reactive for sodalime decarboxylation-

(a) $CH_3 - CH_2 - COOH$

(b) $H_2C = CH - COOH$

(c) $HC \equiv C - COOH$

(1) a > b > c (2) c > b > a

(3) c > a > b (4) a > c > b

42. Given below are two statements: one is labelled as Assertion **(A)** and the other is labelled as Reason **(R).**

Assertion **(A):** Cl^- has lower ionisation energy than S^{2-}.

Reason **(R):** In isoelectronic ions, greater the negative charge lesser will be the ionisation energy.

In the light of above statements choose the most appropriate answer from the following.

(1) Both **A** and **R** are correct and **R** is the correct explanation of **A**.

(2) Both **A** and **R** are correct but **R** is not the correct explanation of **A**.

(3) **A** is correct but **R** is incorrect.

(4) **A** is incorrect but **R** is correct

43. The colour of potassium dichromate is due to:

(1) d-d transition

(2) transition in K^+ ion

(3) ligand-to-metal charge transfer

(4) metal to ligand charge transfer

44. The increasing order of atomic radii of the following Group 13 elements is

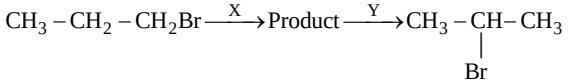
(1) Al < Ga < In < Tl

(2) Ga < Al < In < Tl

(3) Al < In < Ga < Tl

(4) Al < Ga < Tl < In

45. Identify the correct set of reagents/reaction conditions 'X' and 'Y' in the following set of transformations



(1) X = dilute aqueous NaOH, 20°C, Y= HBr/acetic acid, 20°C

(2) X = concentrated alcoholic NaOH, 80°C; Y = HBr/acetic acid, 20°C

(3) X = dilute aqueous NaOH 20°C, Y = $Br_2/CHCl_3$, 0°C

(4) X = concentrated alcoholic NaOH, 80°C; Y = $Br_2/CHCl_3$, 0°C

Integer Type Questions

46. The number of reactions among the following, as written, could describe elementary reaction, is _____.

(i) $Cl_2 + CO \rightarrow COCl_2$; $r = k[Cl_2]^{3/2}[CO]$

(ii) $PCl_3 + Cl_2 \rightarrow PCl_5$; $r = k[PCl_3][Cl_2]$

(iii) $2NO + H_2 \rightarrow N_2O + H_2O$; $r = k[NO]\,[H_2]$

(iv) $2NO + O_2 \rightarrow 2NO_2$; $r = k[NO]^2\,[O_2]$

(v) $NO + O_3 \rightarrow NO_2 + O_2$; $r = k[NO][O_3]$

47. For the reaction $C_2H_6 \rightarrow C_2H_4 + H_2$, the reaction enthalpy Δ_rH =_____ kJ mol^{-1}.
[Given: Bond enthalpies in kJ mol^{-1} : C – C: 347, C = C: 611; C – H: 414, H – H: 436]

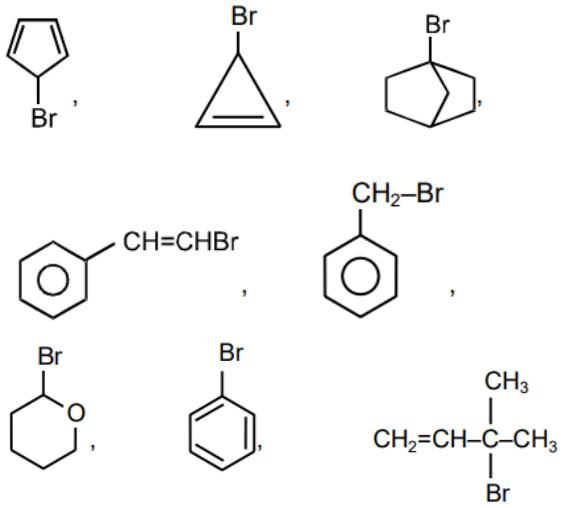
48. Consider the following reaction

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{+2} + 4H_2O, E^0 = 1.51V$$

The quantity of electricity required (in Faraday) to reduce five moles of MnO_4^- is_____
(Integer answer)

49. Observe the following molecules
PCl_5 , BrF_5, PF_5 ,ClF_3 , XeF_4 , XeF_2 , IF_5
The number of molecules having square pyramidal geometry from the above is:

50. How many of these compounds are inert towards nucleophilic substitution by H_2O in aqueous solution?



SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. Let α and β are two solutions of the equation $64 \cdot 9^x - 84 \cdot 12^x + 27 \cdot 16^x = 0$. Then the quadratic equation whose roots are α + β and β – α (β > α) is

(1) $x^2 - 5x + 6 = 0$

(2) $x^2 - 4x + 3 = 0$

(3) $x^2 - 6x + 5 = 0$

(4) $x^2 - 5x + 3 = 0$

52. $\int \frac{3 + 2\cos x}{(2 + 3\cos x)^2} dx$ is equal to

(1) $\left(\frac{\sin x}{3\cos x + 2}\right) + c$ (2) $\left(\frac{2\cos x}{3\sin x + 2}\right) + c$

(3) $\left(\frac{2\cos x}{3\cos x + 2}\right) + c$ (4) $\left(\frac{2\sin x}{3\sin x + 2}\right) + c$

53. Solution of $\log_2(x-2) \le \log_{1/2}(x+1)$ is

(1) $\left[\frac{1-\sqrt{13}}{2}, -1\right)$ (2) $\left(-1, \frac{1+\sqrt{13}}{2}\right]$

(3) $\left(2, \frac{1+\sqrt{13}}{2}\right]$ (4) $\left[\frac{1-\sqrt{13}}{2}, \frac{1+\sqrt{13}}{2}\right]$

54. $\lim\limits_{n\to\infty}\left[\frac{1\cdot n + 2(n-1) + \ldots + n\cdot 1}{1^3 + 2^3 + \ldots + n^3} + 1\right]^n$ is equal to

(1) $\frac{2}{3}$ (2) e^2

(3) $e^{\frac{1}{2}}$ (4) $e^{\frac{2}{3}}$

55. If set $A = \{x : x^3 - 6x^2 + 11x - 6 = 0\}$, $B = \{2, 3, 4\}$ and $C = \{4, 5, 6\}$, then the value of $A \times (B \cap C)$ is

(1) {(–1, 4), (8, 4)}

(2) {(1, 4), (2, 4), (3, 4)}

(3) {(1, 4), (2, 4)}

(4) {(1, 1), (1, 2), (1, 3), (1, 4)}

56. If $2x^y + 3y^x = 20$, then $\frac{dy}{dx}$ at (2, 2) is equal to:

(1) $-\left(\frac{3 + \log_e 8}{2 + \log_e 4}\right)$ (2) $-\left(\frac{2 + \log_e 8}{3 + \log_e 4}\right)$

(3) $-\left(\frac{3 + \log_e 4}{2 + \log_e 4}\right)$ (4) $-\left(\frac{3 + \log_e 16}{4 + \log_e 8}\right)$

57. Among the statements:
(S_1) : $2023^{2022} - 1999^{2022}$ is divisible by 8
(S_2) : 13 $(13)^n$ – 11n – 13 is divisible by 144 for infinitely many $n \in \mathbb{N}$

(1) Only (S_2) is correct

(2) Only (S_1) is correct

(3) Both (S_1) and (S_2) are incorrect

(4) Both (S_1) and (S_2) are correct

58. A line segment AB of length λ moves such that the points A and B remain on the periphery of a circle of radius λ. Then the locus of the point, that divides the line segment AB in the ratio 2 : 3, is a circle of radius:

(1) $\frac{2}{3}\lambda$ (2) $\frac{\sqrt{19}}{7}\lambda$

(3) $\frac{3}{5}\lambda$ (4) $\frac{\sqrt{19}}{5}\lambda$

59. Let $\vec{a}=3\hat{i}+\hat{j}-\hat{k}$ and $\vec{c}=2\hat{i}-3\hat{j}+3\hat{k}$. If $\vec{b}$ is a vector such that $\vec{a}=\vec{b}\times\vec{c}$ and $|\vec{b}|^2=50$, then $\left|72-|\vec{b}+\vec{c}|^2\right|$ is equal to _____.

(1) 67 (2) 68
(3) 66 (4) 5

60. The number of ways in which 7 men and 7 women can sit on a circular table so that no two women sit together is

(1) $7!\cdot 7!$ (2) $7!\cdot 6!$
(3) $(6!)^2$ (4) $7!$

61. In an increasing geometric series, the sum of the second and the sixth term is $\frac{25}{2}$ and the product of the third and fifth term is 25. Then, the sum of 4th, 6th and 8th terms is equal to

(1) 30 (2) 26
(3) 35 (4) 32

62. The complete solution set of

$$\left|\frac{x+1}{x}\right|+|x+1|=\frac{(x+1)^2}{|x|}$$ is

(1) $(0,\infty)$ (2) $(1,\infty)$
(3) $\{-1,1\}$ (4) $(0,\infty)\cup\{-1\}$

63. If $\tan A+\tan B+\tan C=k$, $0<k<10$ and $A+B+C=180°$, then the value of $\tan A\tan B\tan C$, is

(1) $\frac{k}{4}$ (2) $4k$
(3) k^3 (4) k

64. The values of a for which the system of equations

$ax-3y+z=0$
$x+ay+3z=1$
$3x+y+5z=2$

does not have unique solution are

(1) $-1,\ \frac{11}{5}$ (2) $-1,\ \frac{-11}{5}$
(3) $1,\ \frac{-11}{5}$ (4) $1,\ \frac{11}{5}$

65. For a real number x, let $[x]$ denote the greatest integer less than or equal to x. Let $f: R\to R$ be defined as $f(x)=2x+[x]+\sin x\cdot\cos x$ then f is

(1) one-one but not onto
(2) onto but not one-one
(3) both one-one and onto
(4) neither one-one nor onto

66. A water tank in the shape of a right circular cone has a height of 10 feet. The top rim of the tank is a circle with a radius of 4 feet. If water is pumped into the tank at the rate of 2 cubic feet per minute, then the rate of change of the water depth, in feet per minute, when the depth is 5 feet, is

(1) $\frac{1}{\pi}$ (2) $\frac{3}{2\pi}$
(3) $\frac{2}{\pi}$ (4) $\frac{1}{2\pi}$

67. The area of the region bounded by $x=0$; $y=0$; $x=2$; $y=2$, $y\le e^x$ & $y\ge \ln x$ is (in sq. units)

(1) $4\ln 2-2$
(2) $6-4\ln 2$
(3) $2\ln 2-4$
(4) $6-2\ln 2$

68. A sample space consists of the integers 1, 2, 3, 4, ..., 100. The probability of choosing an integer 'k' is proportional to $\ln k$ The conditional probability of choosing the integer 2, given that an even integer is chosen, is

(1) $\frac{\ln 2}{50\ln 2+\ln(50!)}$ (2) $\frac{\ln 2}{\ln 2+\ln(50!)}$
(3) $\frac{\ln 2}{50\ln 2+\ln(50)}$ (4) $\frac{\ln 2}{\ln 2+\ln(50)}$

69. $S_1: R_1=\{(x,y)\in N^2;\ |x|+|y|\le 1\}$ is not transitive
$S_2: R_2=\{(x,y)\in I^2;\ |x|-|y|\le 1\}$ is transitive
$S_3: R_3=\{(x,y)\in R^2;\ |x-y|\le 1\}$ is not transitive
$S_4: R_4:=\{(x,y)\in I^2;\ |x|+|y|\le 1\}$ is not transitive
Then statements are (respectively) True (T)/False(F).

(1) T F T F
(2) F T F T
(3) T T T F
(4) F F T T

70. If $y=y(x), x\in\left(0,\frac{\pi}{2}\right)$ be the solution curve of the differential equation

$$\left(\sin^2 2x\right)\frac{dy}{dx}+\left(8\sin^2 2x+2\sin 4x\right)y$$
$$=2e^{-4x}\left(2\sin 2x+\cos 2x\right),\ \text{with}\ y\left(\frac{\pi}{4}\right)=e^{-\pi},\ \text{then}$$

$y\left(\frac{\pi}{6}\right)$ is equal to

(1) $\frac{2}{\sqrt{3}}e^{-2\pi/3}$ (2) $\frac{2}{\sqrt{3}}e^{2\pi/3}$
(3) $\frac{1}{\sqrt{3}}e^{-2\pi/3}$ (4) $\frac{1}{\sqrt{3}}e^{2\pi/3}$

Integer Type Questions

71. Let $f(x)=\dfrac{\sec^2 x+4\tan x+8}{\sec^2 x}$. If M and m are maximum and minimum value of $f(x)$, then the value of $M + m$ is

72. Let $P=\begin{bmatrix}\frac{\sqrt{3}}{2} & \frac{1}{2}\\ -\frac{1}{2} & \frac{\sqrt{3}}{2}\end{bmatrix}, A=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ and $Q = PAP^T$. If $P^TQ^{2007}P=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$, then $\dfrac{2a+b-3c-4d}{5}$ equal to

73. If the eccentricities of the ellipse $\dfrac{x^2}{4}+\dfrac{y^2}{3}=1$ and the hyperbola $\dfrac{x^2}{64}-\dfrac{y^2}{b^2}=1$ are reciprocals of each other, then b^2 is equal to

74. If the lines $\dfrac{x-1}{2}=\dfrac{2-y}{-3}=\dfrac{z-3}{\alpha}$ and $\dfrac{x-4}{5}=\dfrac{y-1}{2}=\dfrac{z}{\beta}$ intersect, then the magnitude of the minimum value of $8\alpha\beta$ is

75. The number of solutions of $\sin^{-1} x + \sin^{-1}(1 + x) = \cos^{-1} x$ is



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if